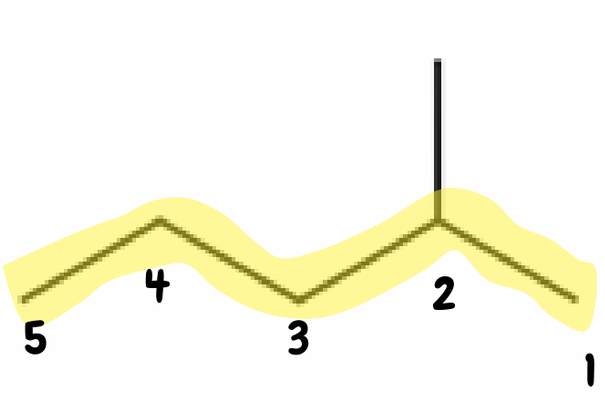
AOI.

Foundation | C_6H_{13}

Numbered Parent



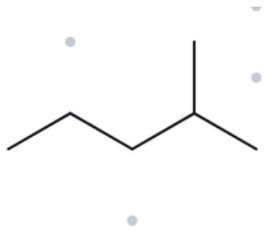
Naming Traile

Parent: longest chain = 5 C → pentane.
Numbering: right → left gives 2; left → right gives 4. First point of difference: 2 < 4, so number from the right.
Substituents: methyl on C2 → 2-methyl.

My Name

2-methylpentane

The Alkane Check



2-methylpentane

Returned Name: 2-methylpentane

Match✓

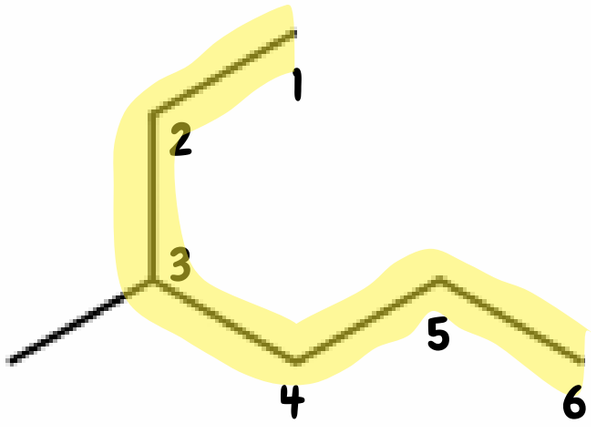
☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order

☑ punctuation ☑ formula counted

AO2.

Foundation | C_7H_{16}

Numbered Parent



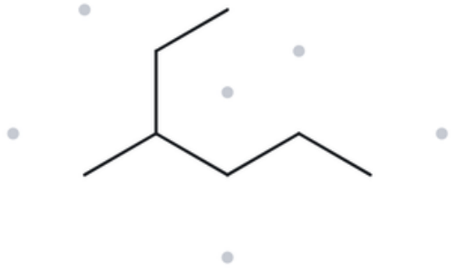
Naming Traile

Parent: longest chain = 6 C → hexane.
Numbering: top → right gives 3; right → top gives 4. First point of difference: 3 < 4, so number from the top.
Substituents: methyl on C3 → 3-methyl.

My Name

3-methylhexane

The Alkane Check



3-methylhexane

Returned Name: 3-methyhexane

Match✓

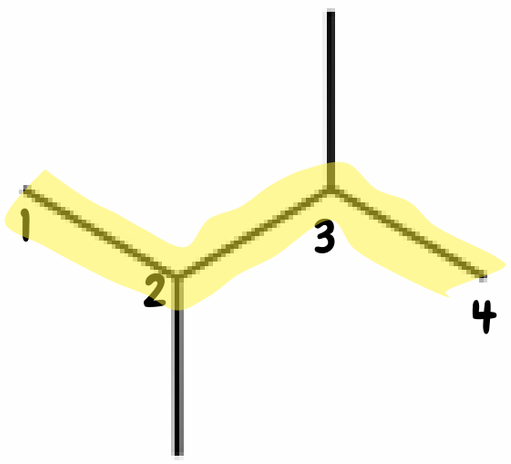
☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order

☑ punctuation ☑ formula counted

AO3.

Foundation | C_6H_{14}

Numbered Parent



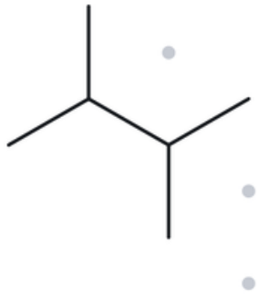
Naming Traile

Parent: longest chain = 4 C → butane.
Numbering: left → right gives 2,3; right → left gives 2,3. Both directions are identical, so number from either side.
Substituents: methyl on C2 and C3 → 2,3-dimethyl

My Name

2,3-dimethylbutane

The Alkane Check



2,3-dimethylbutane

Returned Name: 2,3-dimethybutane

Match✓

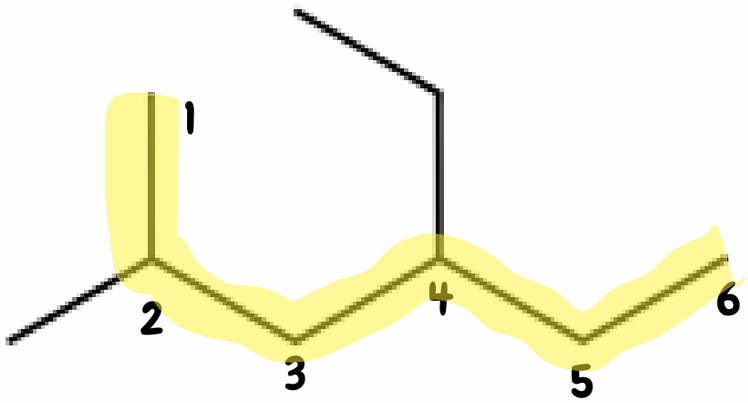
☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order

☑ punctuation ☑ formula counted

AO4.

Developing|C_9H_{20}

Numbered Parent



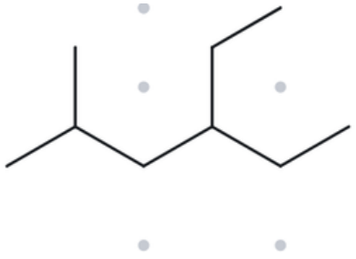
Naming Traile

Parent: longest chain = 6 C → hexane.
Numbering: left → right gives 2,4; right → left gives 3,5. First point of difference: 2 < 3, so number from the left.
Substituents: methyl on C2 and ethyl on C4 → 4-ethyl-2-methyl.

My Name

4-ethyl,2-methylhexane

The Alkane Check



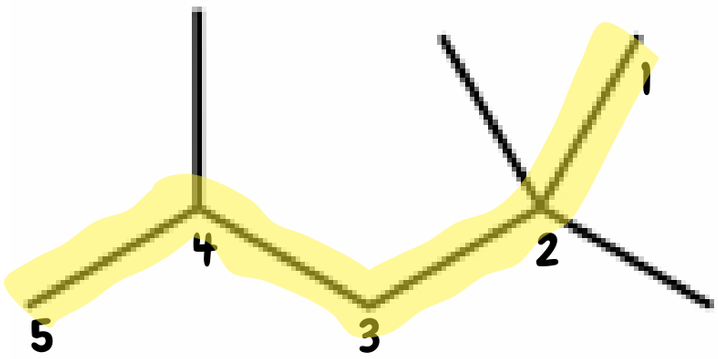
4-ethyl-2-methylhexane
Returned Name: 4-ethyl,2-methylhexane
Match✓

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order
☑ punctuation ☑ formula counted

AO5.

Developing | C_8H_{18}

Numbered Parent



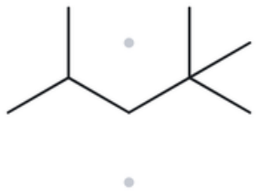
Naming Traile

Parent: longest chain = 5 C → pentane.
Numbering: right → left gives 2,2,4; left → right gives 2,4,4. First point of difference: 2 < 4, so number from the right.
Substituents: methyl on C2, C2 and C4 → 2,2,4-trimethyl.

My Name

2,2,4-trimethylpentane

The Alkane Check



2,2,4-trimethylpentane

Returned Name:2,2,4-trimethylpentane

Match✓

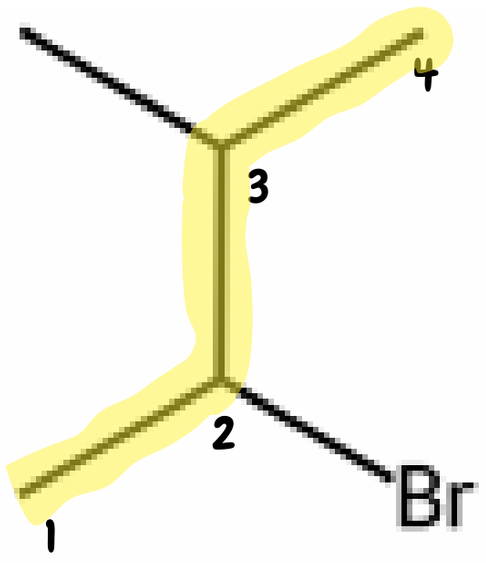
☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order

☑ punctuation ☑ formula counted

AO6.

Developing| $C_5H_{11}Br$

Numbered Parent



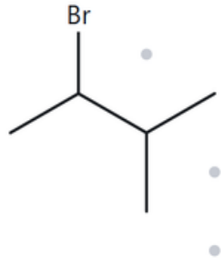
Naming Traile

Parent: longest chain = 4 C → butane.
Numbering: bottom-left → top-right gives 2,3 (2-bromo, 3-methyl); top-right → bottom-left gives 2,3 (3-bromo, 2-methyl). Alphabetical order ("b" before "m") means number from the bottom-left.
Substituents: bromo on C2 and methyl on C3 → 2-bromo-3-methyl.

My Name

2-bromo-3-methylbutane

The Alkane Check



2-bromo-3-methylbutane

Returned Name:2-bromo-3-methylbutane

Match✓

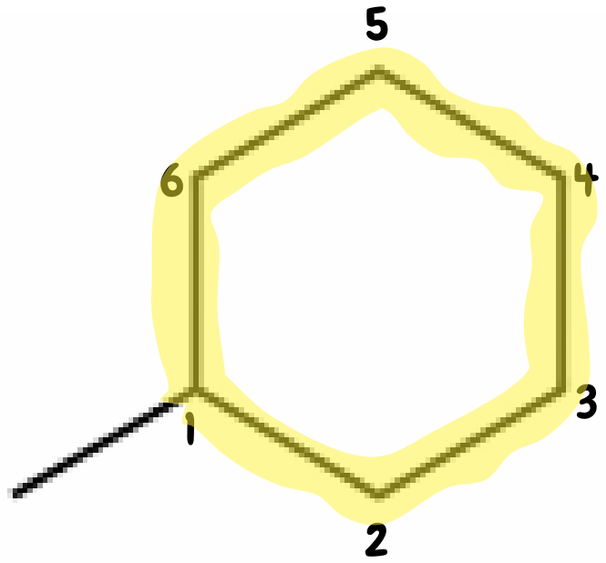
☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order

☑ punctuation ☑ formula counted

AO7.

Developing| C_7H_{14}

Numbered Parent



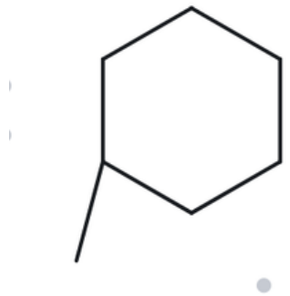
Naming Traile

Parent: 6-carbon ring → cyclohexane.
Numbering:the carbon with the branch must be position 1, so numbering starts directly at that junction.
Substituents: methyl on C1 → methylcyclohexane (the locant number "1" is omitted in the final name for a ring with only one substituent).

My Name

methylcyclohexane

The Alkane Check



methylcyclohexane

Returned Name:methylcyclohexane

Match✓

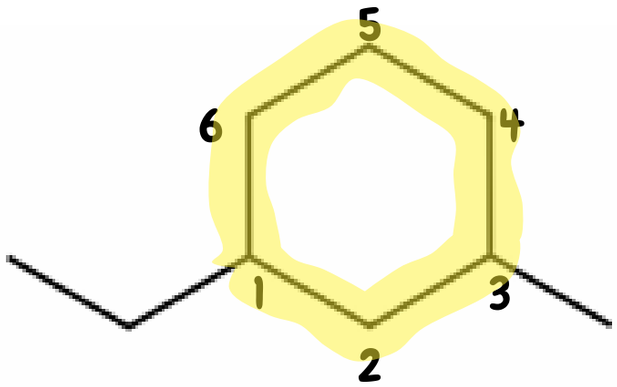
☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order

☑ punctuation ☑ formula counted

AO8.

Challenge | C_9H_{18}

Numbered Parent



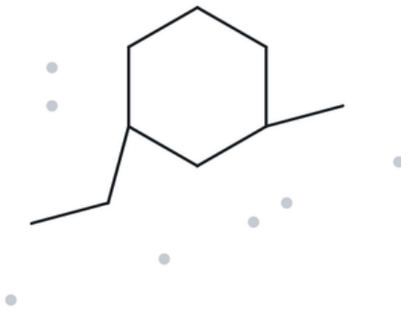
Naming Traile

Parent: 6-carbon ring → cyclohexane.
Numbering: ethyl-to-methyl via the bottom path gives 1,3. Alphabetical order ("e" before "m") means position 1 goes to the ethyl group and we number counter-clockwise toward the methyl group.
Substituents: ethyl on C1 and methyl on C3 → 1-ethyl-3-methylcyclohexane.

My Name

1-ethyl-3-methylcyclohexane

The Alkane Check



1-ethyl-3-methylcyclohexane

Returned Name:1-ethyl-3-methylcyclohexane

Match✓

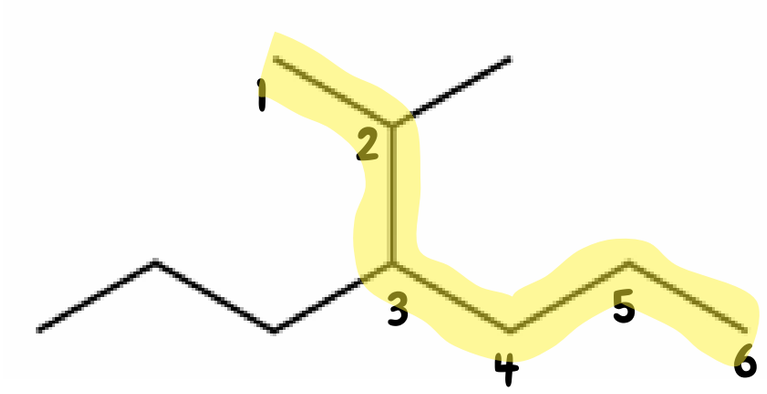
☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order

☑ punctuation ☑ formula counted

AO9.

Challenge | $C_{10}H_{22}$

Numbered Parent



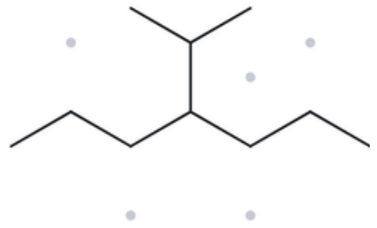
Naming Traile

Parent: longest chain = 6 C → hexane.
Numbering: top-left → bottom-right gives 2,3; bottom-right → top-left gives 4,5. First point of difference: 2 < 4, so number from the top-left branch.
Substituents: methyl on C2 and propyl on C3 → 2-methyl-3-propylhexane.

My Name

2-methyl-3-propylhexane

The Alkane Check



2-methyl-3-propylhexane

Returned Name:2-methyl-3-propylhexane

Match✓

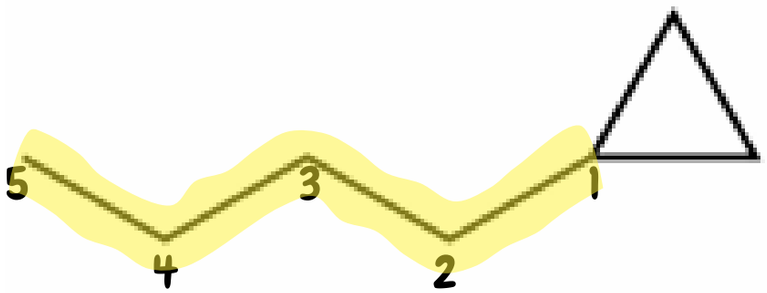
☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order

☑ punctuation ☑ formula counted

A10.

Challenge | C_8H_{16}

Numbered Parent



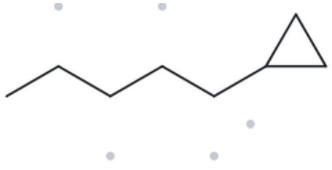
Naming Traile

Parent: longest chain = 5 C → pentane.
Numbering: right → left gives 1; left → right gives 5. First point of difference: 1 < 5, so number from the right.
Substituents: cyclopropyl on C1 → 1-cyclopropylpentane.

My Name

1-cyclopropylpentane

The Alkane Check



pentylcyclopropane

Returned Name:pentylcyclopropane

Note: Mismatched. The ring is the one considered as parent chain.

☑ longest chain confirmed ☑ lowest locants ☑ alphabetical order

☑ punctuation ☑ formula counted

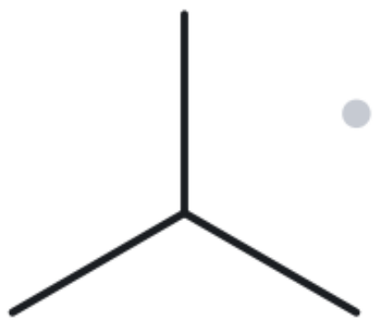
BOI.

Foundation | 2-methylpropane

Name Decoded

Parent: butane → a 3-carbon chain.
Substituents: methyl on C2.
Expected formula: 3 + 1 = 4 C; no rings or double bonds → C_4H_{12}

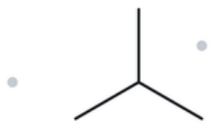
Bond-Line Structure



Formula Check

Counted from the drawing: C_4H_{10}
No rings or double bonds: C_nH_{2n+2} = C_4H_{10}

Back-Check



2-methylpropane

Returned by The Alkane: 2-methylpropane

Same as the given name ✓

☑ parent length matches ☑ substituents position match ☑ formula agrees ☑ back-checked recorded

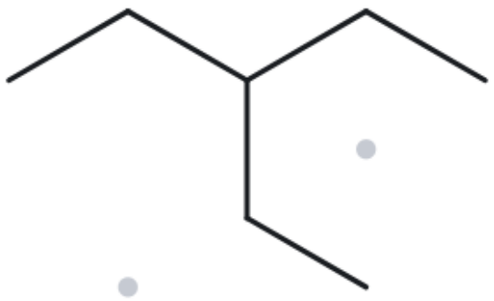
BO2.

Foundation | 3-ethylpentane

Name Decoded

Parent: pentane → a 5-carbon chain.
Substituents: ethyl on C3.
Expected formula: 5 + 2 = 7 C; no rings or double bonds → C_7H_{16}

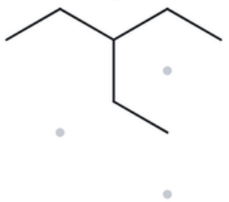
Bond-Line Structure



Formula Check

Counted from the drawing: C_7H_{16}
No rings or double bonds: C_nH_{2n+2} = C_7H_{16}

Back-Check



3-ethylpentane

Returned by The Alkane: 3-ethylpentane

Same as the given name ✓

☑ parent length matches ☑ substituents position match ☑ formula agrees ☑ back-checked recorded

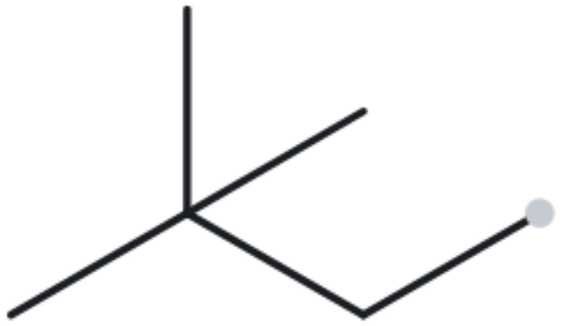
B03.

Foundation | 2,2-dimethylbutane

Name Decoded

Parent: butane → a 4-carbon chain.
Substituents: methyl on C2 and C2.
Expected formula: 4 + 2 = 6 C; no rings or double bonds → C_6H_{14}

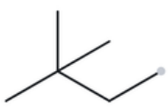
Bond-Line Structure



Formula Check

Counted from the drawing: C_6H_{14}
No rings or double bonds: C_nH_{2n+2} = C_6H_{14}

Back-Check



2,2-dimethylbutane

Returned by The Alkane: 2,2-dimethylbutane

Same as the given name ✓

☑ parent length matches ☑ substituents position match ☑ formula agrees ☑ back-checked recorded

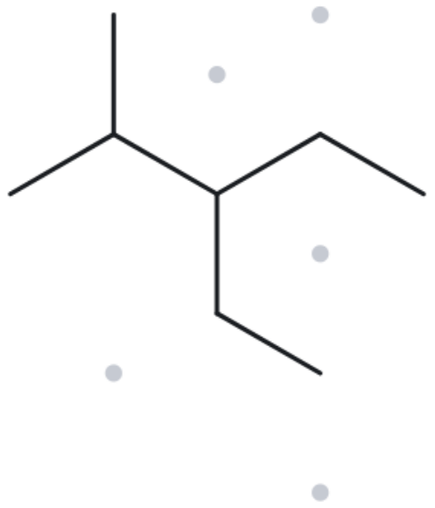
BO4.

Developing | 3-ethyl-2-methylpentane

Name Decoded

Parent: pentane → a 5-carbon chain.
Substituents: methyl on C2; ethyl on C3.
Expected formula: 5 + 1 + 2 = 8 C; no rings or double bonds → C_8H_{18}

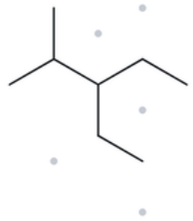
Bond-Line Structure



Formula Check

Counted from the drawing: C_8H_{18}
No rings or double bonds: C_nH_{2n+2} = C_8H_{18}

Back-Check



3-ethyl-2-methylpentane

Returned by The Alkane: 3-ethyl-2-methylpentane
Same as the given name ✓

☑ parent length matches ☑ substituents position match ☑ formula agrees ☑ back-checked recorded

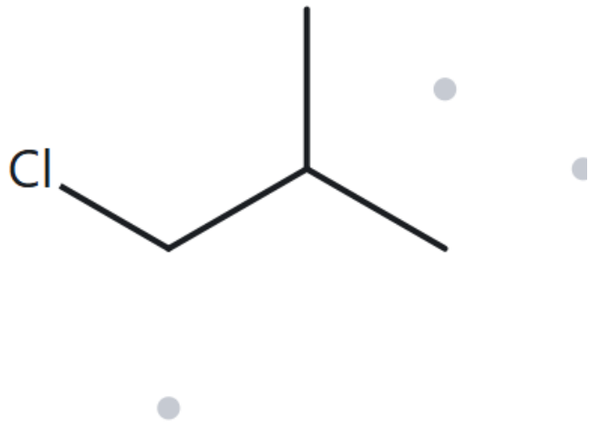
BO5.

Developing | 1-chloro-2-methylpropane

Name Decoded

Parent: propane → a 3-carbon chain.
Substituents: chloro on C1; methyl on C2.
Expected formula: 3 + 1 = 4 C; no rings or double bonds → C_4H_{10}; Cl replaces one H→ C_4H_9Cl

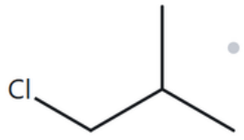
Bond-Line Structure



Formula Check

Counted from the drawing: C_4H_9Cl
No rings or double bonds: C_nH_{2n+2} = C_4H_{10}; minus one H for Cl = C_4H_9Cl

Back-Check



1-chloro-2-methylpropa

Returned by The Alkane: 1-chloro-2-methylpropane

Same as the given name ✓

☑ parent length matches ☑ substituents position match ☑ formula agrees ☑ back-checked recorded

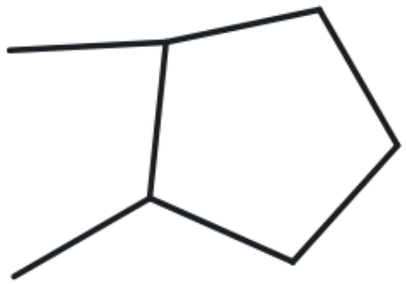
BO6.

Developing | 1,2-dimethylcyclopentane

Name Decoded

Parent: cyclopentane → a 5-carbon ring.
Substituents: methyl on C1 and C2.
Expected formula: 5 + 2 = 7 C;
one ring → C_7H_{14}

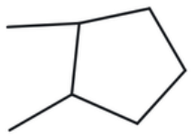
Bond-Line Structure



Formula Check

Counted from the drawing: C_7H_{14}
One Ring: C_nH_{2n} = C_7H_{14}

Back-Check



1,2-dimethylcyclopentane

Returned by The Alkane: 1,2-dimethylcyclopentane
Same as the given name ✓

☑ parent length matches ☑ substituents position match ☑ formula agrees ☑ back-checked recorded

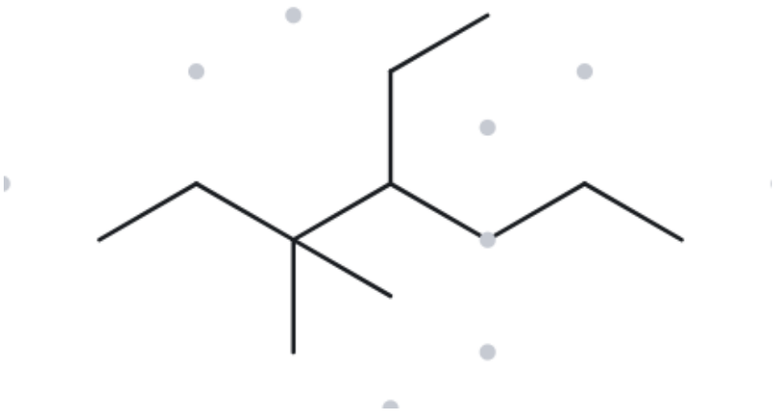
BO7.

Developing | 4-ethyl-3,3-dimethylheptane

Name Decoded

Parent: heptane → a 7-carbon chain.
Substituents: methyl on C3 and C3; ethyl on C4.
Expected formula: 7 + 2 + 2 = 11 C; no rings or double bonds → $C_{11}H_{24}$

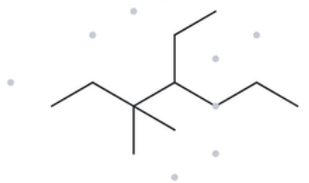
Bond-Line Structure



Formula Check

Counted from the drawing: $C_{11}H_{24}$
No rings or double bonds: C_nH_{2n+2} = $C_{11}H_{24}$

Back-Check



4-ethyl-3,3-dimethylheptane

Returned by The Alkane: 4-ethyl-3,3-dimethylheptane
Same as the given name ✓

☑ parent length matches ☑ substituents position match ☑ formula agrees ☑ back-checked recorded

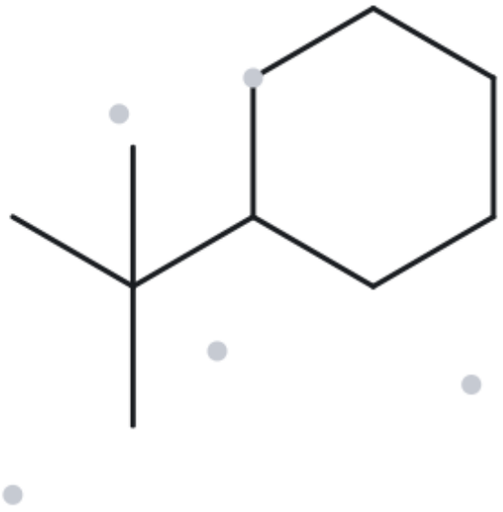
B08.

Challenge | tert-butylcyclohexane

Name Decoded

Parent: cyclohexane → a 6-carbon ring.
Substituents: tert-butyl on C1.
Expected formula: 6 + 4 = 10 C; one ring → $C_{10}H_{20}$

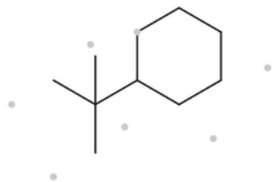
Bond-Line Structure



Formula Check

Counted from the drawing: $C_{10}H_{20}$
No rings or double bonds: C_nH_{2n} = $C_{10}H_{20}$

Back-Check



(1,1-dimethylethyl)cyclohexane

Returned by The Alkane: (1,1-dimethylethyl)cyclohexane

The website displays the formal, systematic IUPAC name instead of the common "tert-butyl" nickname.

☑ parent length matches ☑ substituents position match ☑ formula agrees ☑ back-checked recorded

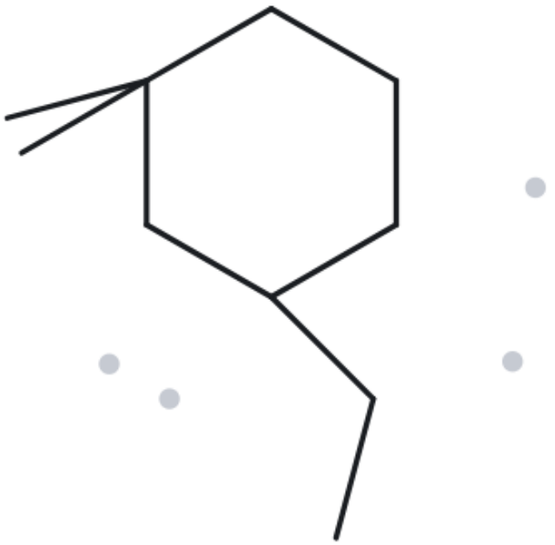
BO9.

Challenge | 3-ethyl-1,1-dimethylcyclohexane

Name Decoded

Parent: cyclohexane → a 6-carbon ring.
Substituents: methyl on C1 and C1; ethyl on C3.
Expected formula: 6 + 2 + 2 = 10 C; one ring → $C_{10}H_{20}$

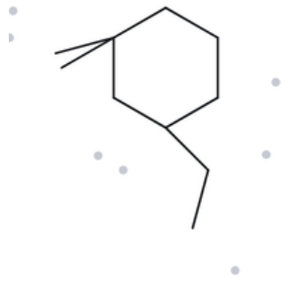
Bond-Line Structure



Formula Check

Counted from the drawing: $C_{10}H_{20}$
No rings or double bonds: C_nH_{2n} = $C_{10}H_{20}$

Back-Check



3-ethyl-1,1-dimethylcyclohexane

Returned by The Alkane: 3-ethyl-1,1-dimethylcyclohexane

Same as the given name ✓

☑ parent length matches ☑ substituents position match ☑ formula agrees ☑ back-checked recorded

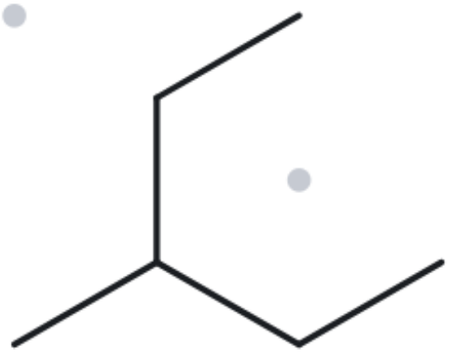
BIO.

Challenge | 2-ethylbutane

Name Decoded

Parent: butane → a 4-carbon chain.
Substituents: ethyl on C2.
Expected formula: 4 + 2 = 6 C; no rings or double bonds → C_6H_{14}

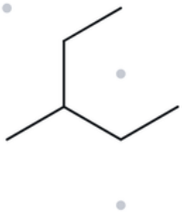
Bond-Line Structure



Formula Check

Counted from the drawing: C_6H_{14}
No rings or double bonds: C_nH_{2n+2} = C_6H_{14}

Back-Check



3-methylpentane

Returned by The Alkane: 3-methylpentane

Since the initial name is incorrect, it misidentifies the parent chain and branches, causing the expected structure to mismatch

☑ parent length matches ☑ substituents position match ☑ formula agrees ☑ back-checked recorded